Section 21.4 Hardcopy option

CONTENTS

1. HARDCOPY INTRODUCTION

With release 7, one Point to point (P2P) HC printer or one Dicom HC printer can be connected to the Gyroscan Intera system The P2P connection (like usual before R7) is established with a Control line and a DATA line between the HCU and the IIM, which is located in the Backend Miscellaneous box. The Dicom printer can be connected to the local MR network if it is dedicated to the MR system or can be connected somewhere to the hospital network if it is also used by other modalities.

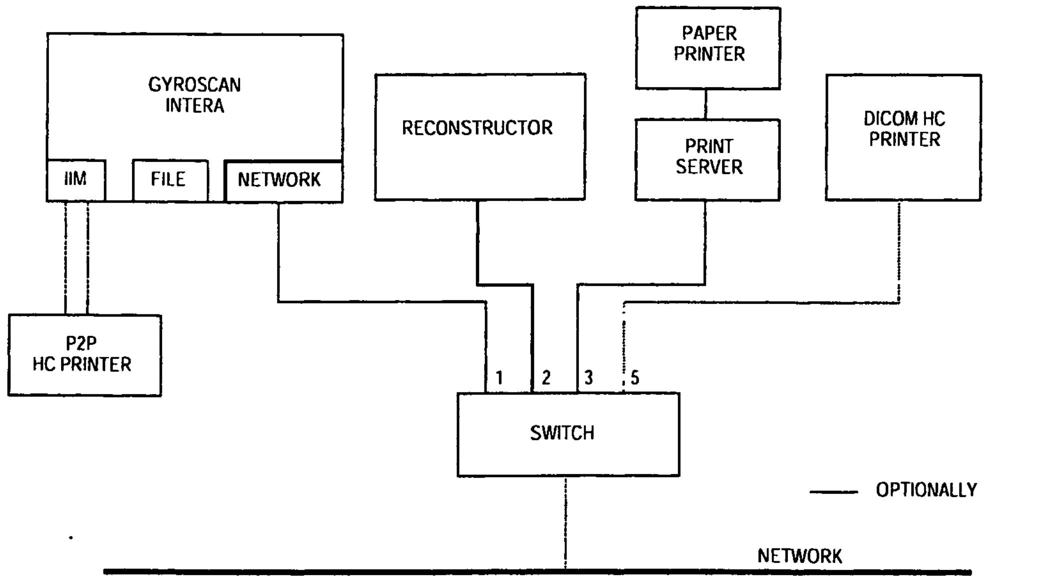

Figure 1 Hardcopy location

The next table gives an overview of the certified Dicom HC printers, and the resource (default settings) file selection. This info is also available at the on-line info (help) in Gyrotool.

DICOM:

Camera type	**Value**
Agfa MG 3000	AG_MG3000
Agfa Drystar 2000	AG_DS2000
Agfa Drystar 3000	AG_DS3000
Agfa LR 3300 / LR 5500	AG_LR3300
Fuji Dicom gateway PS551	FU_PS551
Imation (3M) Dryview 8300	IM_DV8300
Imation (3M) Pacslink	IM_PACSLINK
Kodak MLP 190	KD_MLP190
Sterling Contact 400	ST_CONT400
Sterling SIJ400	ST_SIJ400

The next table gives an overview of the certified P2P HC printers, and the resource file selection This info is also available at the on-line info (help) in Gyrotool

Point to Point:

Camera type	**Value**
Agfa MCL (+MIN)	AG_MCL
Agfa MG 3000	AG_MG3000
Agfa Drystar 2000	AG_DS2000
Agfa Drystar 3000	AG_DS3000
Agfa LR 3300 / LR 5500	AG_LR3300
Fuji FL-IM-3543 + MF300	FU_IM3543
Fuji FL-IM-3535 + MF300	FU_IM3535
Fuji FL-IM-2636 + MF300	FU_IM2636
Fuji FL-IM-D + MF300	FU_IM_D
Imation (3M) P831	IM_P831
Imation (3M) M952 (+ MMU)	IM_M952
Imation (3M) M959 XL	IM_M959
Imation (3M) M969 HQ (S)	IM_M969
Imation (3M) Dryview 8100	IM_DV8100
Imation (3M) Dryview 8300	IM_DV8300
Imation (3M) Dryview 8500	IM_DV8500
Imation (3M) Dryview 8700	IM_DV8700
Imation (3M) HQT	IM_HQT
Kodak KELP 100 (+KEIM)	KD_K100
Kodak KELP 100 XLP	KD_K100XLP
Kodak KELP 1120	KD_K1120
Kodak KELP 2180	KD_K2180
Konica LI10A	KO_LI10A
Konica LI21	KO_LI21
Sterling LLI + LINX	ST_LLI
Sterling LP300 + LINX	ST_LP300
Sterling LP400 + LINX	ST_LP400

All software configurations for hardcopy are done under user Gyrotool: Modify system configuration. The information is stored in the file GYRO$SITE: gyrofig.idx

Hardcopy Unit installation can be done during initial software installation or afterwards following the procedure in chapter 3:

2. HARDWARE INSTALLATION

2.1. P2P HC PRINTER

1. The power for a hardcopy printer (HC printer) can be taken directly from the hospital mains. The signal cables between the HC printer and the MR system are isolated to prevent earth loops. Only be sure in this case that the screening of the cables between the MR system and HC printer are connected at only one side, preferable at the MR system.

NOTE

The GYROSCAN system does not support Remote Control with use of a Hardcopy Unit keypad.

2. The following cables (1.5 m) are delivered with the system and can be used to connect to the HC printer cables according to the following layout. They can also be used as testcables for the HC testbox.

IIM: BE X15 - Image Data - BF Hardcopy Unit
IIM: BE X16 - Control Data - BF Hardcopy Unit

CAUTION

Be sure to make the correct connection for the DATA and CONTROL cables.
Wrong connections may destroy the IIM hardcopy interface !!!

Control Data cable.

Pin configuration of the host control cables is as follows:

	BF	BE-X16	BEIB-X2
Control Send Data -	3	22	2
Control Send Data +	21	4	1
Control Receive Data -	2	24	5
Control Receive Data +	20	6	4
Ground	7	19	3

Image Data cable.

The data cable between IIM and HC printer is 1:1 configured.

3. Switching on.

For some HC printers it can be necessary that it have to be switched on before you boot-up the host computer. Only then you can be sure that the communication between the HC printer and the Gyroscan is correct.

2.2. DICOM HC PRINTER

The supplier of the DICOM printer must do the network connections of the DICOM printer.
The DICOM printer can be connected directly to the hospital network if it is also used for other modalities in the hospital . (See also drawing Z1-9.6 in the console section).
If it is a dedicated MR DICOM printer it can be connected to one of the reserved positions on the Network Switch delivered with the MR system. In case of network communication problems, use can be made of a Dicom Protocol Monitor (see Newsletter ICS-011). See Section 1 of the SMI manual for the codenumbers of this tool.

3. SOFTWARE CONFIGURATION

3.1. SETUP P2P HC PRINTER

3.1.1. SELECT HC PRINTER

1. Login on the host with user **GYROTOOL**
2. Select: Modify System Configuration
3. Select: Console configuration
4. Select: Copy facilities
5. Set "P2P camera" value according to installed P2P HC printer.
   If a P2P selection is made, leave the Dicom camera setting to None, otherwise it will not accept the selection.
6. Select: Proceed
7. Select: Cancel
8. Select: Exit

After the SW configuration, the detached processes and the GYROSCAN application need to be restarted, so: Login with: **GYRORESTART** on the Console.

After the restart it is possible to hardcopy within the configured system.

3.2. SETUP DICOM HC PRINTER

3.2.1. SELECT DICOM HC PRINTER

1. Login on the host with user **GYROTOOL**
2. Select: Modify System Configuration
3. Select: Console configuration
4. Select: Copy facilities
5.

NOTE

If a DICOM selection is made, leave the P2P camera setting to None, otherwise it will not accept the selection

6. Set "DICOM camera" value according to installed DICOM HC printer.
   In this example the AGFA AG_MG3000 DICOM printer is selected.
7. Select: <Proceed>
8. Select: <Cancel>
9. Select: Exit

3.2.2. CONFIGURE DICOM HC PRINTER

In this example the AGFA AG_MG3000 DICOM HC printer is selected.

1. Login on the host computer with user **system** + **<password>**.
2. Select: `DICOM management`
3. Select: `Configure DICOM Printer`

4. The system responds with:
`Please enter ISGDICOMagfa.scu_ae_title: []:`
Press <**Return**> in case a single provider DICOM printer is used on the network.

NOTE

If the DICOM printer supports provider functionality for more than one user (MRI 1, MRI 2, etc), a scu_ae_tiltle must be given on each MR system e.g. MR1.

5. The system responds with:
`Please enter ISGDICOMagfa.scp_ae_title [MCL]:`
Press <**Return**> or a new Service Class Provider Application Entity title, according the DICOM printer setup. In this example the default is selected.
6. The system responds with:
`Please enter ISGDICOMagfa.port [104]:`
Enter the portnumber to where the DICOM printer is sensing. In this example the default is selected.
7. The system responds with:
`Enter internet address for dicom_agfa []:`
8. Enter IP address of the DICOM printer: **xxx.xxx.xxx.xxx**
9. Now the system displays the values entered.

```
    LOCAL database

Host address              Host name

xxx.xxx.xxx.xxx           DICOM_AGFA, dicom_agfa
```

10. Press <**Return**> for `DICOM Management Menu`
11. Select: `Exit to Gyroscan Manager menu`
12. Select: **Logout** and confirm by clicking **yes**.

NOTE

*Some processes should be restarted, so a GYRORESTART **must** be executed before you can use the configured DICOM printer.*

4. CHECKING HARD COPY UNIT FUNCTIONALITY

4.1. PROCEDURE TO CHECK HARDCOPY FUNCTIONALITY

To check the Hardcopy functionality you have to make a print of the SMPTE test pattern under Gyroscan and check if the hardcopy contrast steps are identical to those on the monitor screen.
In the database special test-images are available.

Procedure to view the test images:

1. Login: **Gyroscan** (wait for the start up process)
2. Select: System (on the upper left-hand side menu bar)
3. Select: Service

   Now 3 test-images and 5 clinical images become available with patient name: 'Service'

   With the next/previous image (scroll bar) selection it is possible to scroll through the images.

4. Select: HC Screen to print a image to the HC printer
5. Check if the brightness and contrast of the film has the same quality as the image on the monitor.
6. If necessary, check the film processor and/or hardcopy unit by making a sensitogram. This is the responsibility of the hospital (HCU manufacturer).

WARNING

The patient name while viewing the test images is : Service.
To prevent any problems with this patient name appearing in regular scans,
it is not allowed to perform any scanning actions after working with this 'Service' patient.
You have to EXIT Gyroscan and afterwards start it again before continuing with regular scanning.

7. Select: System
8. Select: Exit

If the Hardcopy P2P interface does not work, the PMS hardware can be tested with the PMS Hardcopy test box (12NC: 8122 102 22601). This testbox can be connected in place of the HCU and tests real time both Host control and Digital image data transfer.
In the TSW, TIIM software is provided to work with the test box.
Refer to manual: SERVICE MANUAL UNIT, HARDCOPY UNIT TESTBOX (12NC 4522 983 61802) or DSS/SMARTEXT, how to work with the test box (see also paragraph 2.2.2).
Also the Hardcopy printer test tool (12NC 4522 980 38561) can be used to analyze the communication between the IIM and HCU. See newsletter MR-093 for more information.

4.2. CHECKING THE P2P CONNECTIONS

4.2.1. POWER-ON SELFTEST OF THE IIM-CPU

On power-up the boot-ROM on the CPU board of the IIM runs a self-test.
Result of the Self-test is shown on LEDs and reported to the Host computer.
The status of the LEDs can be seen when the front door of the Operator Equipment cabinet is opened.
The LEDs can be seen in the Backend Miscellaneous box.
There are 6 LEDs ·

P	1	3
G	0	2

LED P and G are green
LED 0 to 3 are red

The "P" LED indicates the presence of power on the IIM CPU board, this LED is hard wired.
All other LEDs are software controlled.

The RAM test (LED 2) may take up to 1 minute, depending on the amount of IIM-CPU memory.
The green "G" LED will flash on/off after the selftest. If the IIM-CPU RAM memory has been loaded from the host it will be on continuously (from SW R4.5 onwards).

Other LED status codes are:

Action	LED P	LED G	LED 3	LED 2	LED 1	LED 0
Power-off	Off	Off	Off	Off	Off	Off
I960 test	On	Off	Off	Off	Off	Off
Eprom test	On	Off	On	Off	Off	Off
Timer 0 test	On	Off	Off	On	On	On
Timer 1 test	On	Off	Off	On	On	Off
Timer 2 test	On	Off	Off	On	Off	On
RAM test	On	Off	Off	On	Off	Off
SCSI test	On	Off	Off	Off	On	On
UART A test	On	Off	Off	Off	On	Off
UART B test	On	Off	Off	Off	Off	On
READY	On	On/Off	Off	Off	Off	Off

If one of the red LEDs is on after power-up, refer to corrective action for IIM in DSS.

4.2.2. CHECKING THE P2P INTERFACE WITH THE TESTBOX

The Gyroscan hardcopy interface can be tested with the testbox, by running the TSW TIIM test.
This TIIM test uses the 3M protocol, which is automatically selected by TIIM independent of the ASW protocol selection. So if the TIIM test runs successfully and the Hardcopy Unit is still not working, check that you have configured the correct ASW protocol for your Hardcopy Unit. See paragraph 1.2 of this Section, for selecting the protocol of the Hardcopy Unit connected.

Procedure to check the system hardcopy interface:

1. Switch off the Laser Hardcopy Unit. Disconnect the Laser Hardcopy Unit and connect the testbox instead (use the Laser Hardcopy Unit's cables)

2. Check the switches on the rear of the testbox :
   -Even parity (for Image data)
   -RS422 (for host control)

3. Switch-on the power supply of the testbox
   The default settings of the testbox after switch-on (or reset) are as follows:
   -The testbox goes to auto mode,
   -Parity checks (parity sel. switch => even/odd) for the digital interface are switched on,
   -The serial port is set to 1200 baud,
   -The serial port uses even parity.

4 Check on the IIM box in the OC Expansion Console if the 2 green LED's are on. It is possible that on of these LEDs is flashing when the firmware is not yet downloaded.

5 Login under user TSW and run TIIM.

6. To check if all I/O boards specified in the hardware configuration file are present in the IIM, and to download the software into the CPU RAM memory run the following test:

   Enter: **RUN FUNCTIONAL AUTOMATIC** (will run configuration and CPU tests)

   The automatic test will give you an overview of boards found in the IIM box.
   Now the flashing green LED will be on continuously (from S.W. R4.5 onwards).

   Test should end with: "FUNCTIONAL AUTOMATIC" test succeeded.

Check the Serial and Parallel Hardcopy links

The Parallel image link test, tests the parallel image link and the host control link. It will allocate an image of 32x32 pixels (1 byte per pixel) and send it 100 times to the Hardcopy Unit testbox. The Hardcopy Unit testbox must be in 'automatic' and 'host' mode during this test. The Hardcopy Unit testbox will check all received data. If any error is detected a timeout will be generated in the TIIM software. The cause of the error can be retrieved from the Hardcopy Unit testbox.
Because the test is executed with an automatic acquire command, the serial communication channel is tested as well as the parallel image link.

7. To run the test:

On the hardcopy test box:
    Press soft key: HOST to enter host mode
On the OC:
    Enter: **RUN FUNCTIONAL IIMO PHC_B0 IMAGE_LINK**
    Press: **<RETURN>**

On both the TSW display and the testbox the number of images are counted up until 100.

LED action on the testbox is as follows:

DB0	: blinking
PARITY	. blinking
MODE	: ON
HOST 422 IN	: blinking
HOST 232 IN	: ON
HOST OUT	: blinking

The test will end with the request to disconnect the hardcopy test box.

On the OC:
    Press: <RETURN>
    Test reports: 'IIMO PHC_B0 IMAGE_LINK' test succeeded

Serial Link test only:

If nothing has happened the testbox will time-out after a while. In that case check the logging of the testbox by pressing the LOG softkey from the main menu. If you see commands like : RQS or RES or AQI or PRI and responses of the testbox like PAS or STA,1,RDY, the host control is functioning correctly. If not, no data in buffer will be displayed on the testbox. In that case check the serial connection. To eliminate a cable problem, use the hardcopy cables delivered with the testbox. You can also use a special loopback connector (4522 131 54581) for the serial link and then run the 'serial_loopback' test.

8. On the hardcopy test box:
    Press soft key: HOST to enter HOST mode
On the OC:
    Enter: **RUN FUNCTIONAL IIMO PHC_B0 SERIAL_LOOPBACK**
    Press: **<RETURN>**

Follow the instructions on the screen to check the complete serial link.

5. PRINT TO FILE OPTION

The print to file option creates a Portable Pixel Map file in a posix directory.
Print to file can be switched on via Gyrotool:
Select: Modify system configuration
Select: Console configuration
Select: Copy facilities
Select: Disk printer and toggle it to enable.

The procedure on how to get this file into a gyroscan DCL directory is as follows.

Login: **gyrotest (+password)**

```
$posix
psx> cd /usr/tmp
psx> ls -l
total 9730
-rw-rw-rw-   1 gyrotest system   3932177 Jun 16 12:54 ISGppm.1.0
lrwxrwxrwx   1 system   system        31 Jun 16 12:04 hardcopy ->
-rw-rw-rw-   1 gyrotest system         0 Jun 17 08:01 local7.info
lrwxrwxrwx   1 system   system        31 Jun 16 12:04 prserver ->
-rw-rw----   1 gyrotest system   1048593 Jun 16 12:53 screen_dump

psx> cp ISG* hardcopy
psx> exit
$ do hctmp
DOWN: Moved down to SYS$SPECIFIC:[GYROSCAN.HCTMP].
$ dir

Directory SYS$SPECIFIC:[GYROSCAN.HCTMP]

HC.FRAMES;1          HC.JOBS;1            HC.QUEUE;1
ISGPPM.1;1

Total of 4 files.
$ ren isgppm.1;1 photo1.ppm
%RENAME-I-RENAMED, SYS$SPECIFIC:[GYROSCAN.HCTMP]ISGPPM.1;1 renamed to
SYS$SPECIFIC:[GYROSCAN.HCTMP]PHOTO1.PPM;1
$ dir/size

Directory SYS$SPECIFIC:[GYROSCAN.HCTMP]
HC.FRAMES;1             16400
HC.JOBS;1                  16
HC.QUEUE;1                 16
PHOTO1.PPM;1             7681

Total of 4 files, 24113 blocks.
$
```

Now this PPM file can be downloaded with ProComm using the Kermit protocol or the ZMODEM protocol.
Make sure you use a binary transfer.

On your PC the picture can be viewed with a viewer which support Portable Pixel Map (PPM) files, e.g. Paint Shop Pro.

After transfer delete the files on the Gyroscan system.